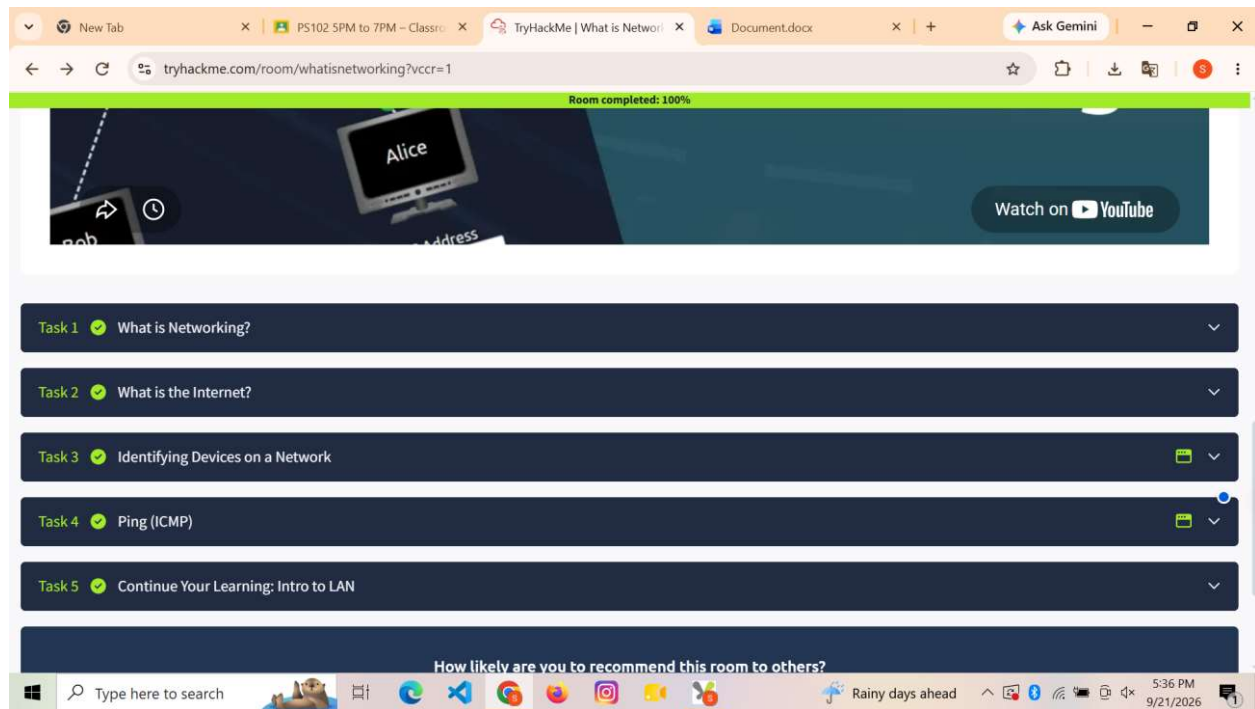


What is networking?

SCREENSHOTS



1.what is networking?

- Explain network, client, server, packet, IP address, MAC address and ping.

1.1 A **network** is a group of devices connected together so they can communicate and share data.

1.2 A **client** is a device or program that **requests a service or data** from another device.

1.3 A **server** is a computer or system that **provides services, data, or resources** to clients.

1.4 A **packet** is a small unit of data sent across a network.

1.5 An **IP address** is a logical address used to identify a device/interface on a network and route IP traffic to it.

1.6 **Ping** is a network utility used to test whether a destination is reachable and to measure **round-trip time (RTT)**.

2. • Explain the difference between an IP address and a MAC address.

MAC addresses = local network delivery.

IP addresses = IP communication and routing between networks.

An **IP address** is a logical Layer-3 address used for IP communication and routing, while a **MAC address** is a Layer-2 identifier used primarily for communication on the local network.

3. • Explain how two computers communicate on a network.

Computer A wants to send a message to Computer B.

1. **Computer A creates the message**
→ "Hello"
2. **Message is broken into small packets**
→ Like small pieces of the message.
3. **IP address finds the destination**
→ A sends the packets to B's IP address.
4. **MAC address helps deliver it locally**
→ It helps the local network know which device should receive it.
5. **Switch/router sends the data**
→ The network devices forward the packets toward B.
6. **Computer B receives the packets**
→ It puts them together and gets the original message.

4. Include at least one simple real-world example.

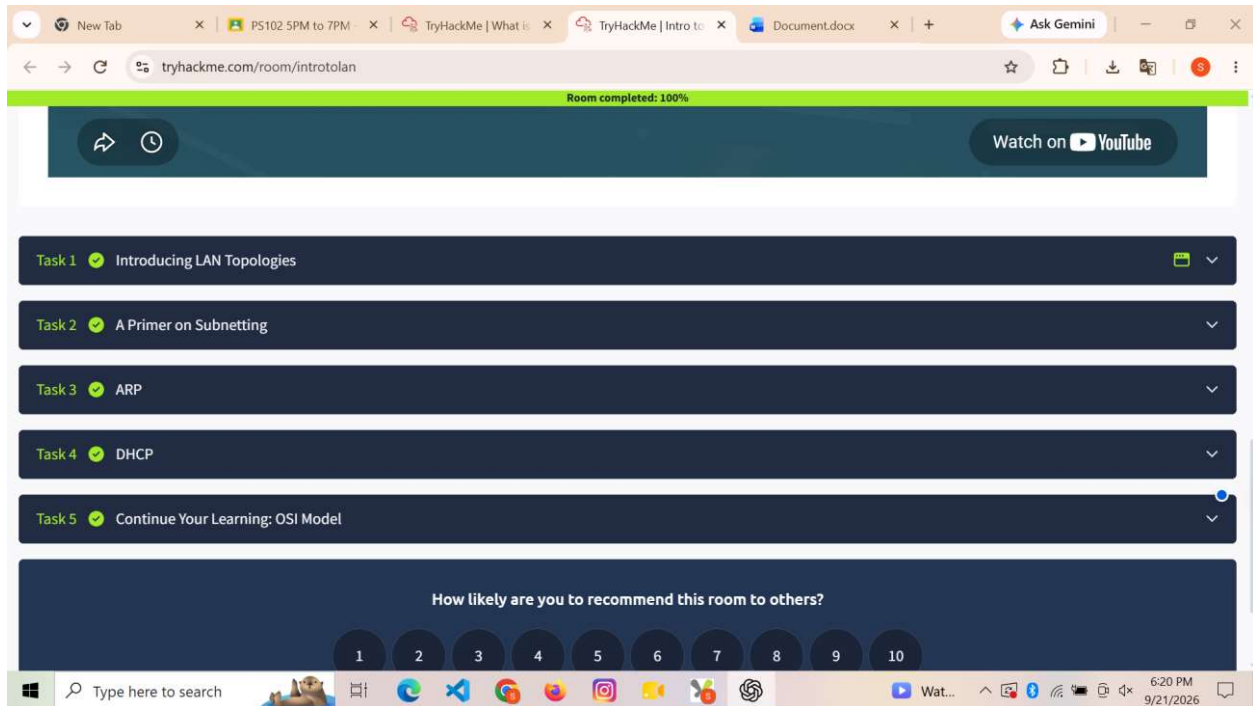
laptop sends a file to another laptop using Wi-Fi:

Laptop A → Wi-Fi Router → Laptop B

The file is broken into packets, sent through the network, and received by Laptop B, where the packets are put back together to form the file.

Introduction to LAN

SCREENSHOT



1. • Define LAN.

A **LAN** is a network that connects computers and other devices within a **small geographical area**, such as a **home, office, school, or college**.

2• Explain the purpose of a switch.

Purpose:

A **switch** is a networking device that **connects multiple devices** (computers, printers, servers, etc.) on the same **LAN** and sends data to the **correct device**.

3• Identify common components of a LAN.

A **LAN (Local Area Network)** usually contains these components:

1. **Computers/Devices** – PCs, laptops, printers, servers, etc.
2. **Switch** – Connects multiple devices within the LAN.
3. **Router** – Connects the LAN to other networks, such as the Internet.
4. **Network cables** – Connect devices using Ethernet cables.
5. **Wireless Access Point (AP)** – Provides Wi-Fi connectivity.

6. **Network Interface Card (NIC)** – Allows a device to connect to the network.

7. **IP Address** – Identifies each device on the network.

4• Explain network topology and name a topology discussed in the room.

there has been experimentation and implementation of various network designs.

Star Topology

The main premise of a star topology is that devices are individually connected via a central networking device such as a switch or hub.

Bus Topology

This type of connection relies upon a single connection which is known as a backbone cable. This type of topology is similar to the leaf off of a tree in the sense that devices (leaves) stem from where the branches are on this cable.

Ring Topology

The ring topology (also known as token topology) boasts some similarities. Devices such as computers are connected directly to each other to form a loop, meaning that there is little cabling required and less dependence on dedicated hardware such as within a star topology.

5• Give one real-world example of a LAN.

Example: College Computer Lab

A college computer lab may have **20 computers connected to a switch**. The computers can communicate with each other and share resources like **printers and files**.

TRYHACKME ROOMS QUESTIONS

1.What does LAN stand for?

Local area network

2. What is the verb given to the job that Routers perform?

Routing

3.What device is used to centrally connect multiple devices on the local network and transmit data to the correct location?

switch

4.What topology is cost-efficient to set up?

Bus topology

5.What topology is expensive to set up and maintain?

Star topology

6.Complete the interactive lab attached to this task. What is the flag given at the end?

THM{TOPOLOGY_FLAWS}

7. What is the technical term for dividing a network into smaller pieces?

subnetting

8.How many **bits** are in a subnet mask?

32

9.What is the range of a section (octet) of a subnet mask?

0-255

10.What address is used to identify the start of a network?

Network address

11.What address is used to identify devices within a network?

Host address

12.What is the name used to identify the device responsible for sending data to another network?

Default gateway

13.What does ARP stand for?

Address resolution protocol

14.What category of ARP Packet asks a device whether or not it has a specific IP address?

Request

15. What address is used as a physical identifier for a device on a network?

Mac address

16. What address is used as a logical identifier for a device on a network?

Ip address

17. What type of DHCP packet is used by a device to **retrieve an IP address**?

DHCP discover

18. What type of DHCP packet does a device **send once it has been offered an IP address** by the DHCP server?

DHCP request

19. Finally, **what is the last** DHCP packet that is sent to a device from a DHCP server?

DHCP ACK

